AI Tools for Research

PART A · SECTION 1

The AI Landscape

Evolution of AI — Six Stages

1

Rule-Based AI

Hand-written IF–THEN rules; no learning; brittle (e.g., MYCIN)

2

Machine Learning

Algorithms learn from labelled examples (e.g., spam filter)

3

Deep Learning

Multi-layer networks learn features from raw data (e.g., tumour detection)

4

Generative AI

Models create new text, images or code

5

Agentic AI

LLM + tools in a loop: plan → act → observe → revise

6

Human–AI Collaboration

Humans set purpose and judge; AI contributes speed and scale

The Seven Forces Shaping AI

Memory aid: **M-R-A-S-S-P-T** — these forces reinforce one another.

M

Multimodal Foundation Models

R

Reasoning-Oriented Models

A

Agentic AI

S

Small & Efficient Models

S

AI for Science & Engineering

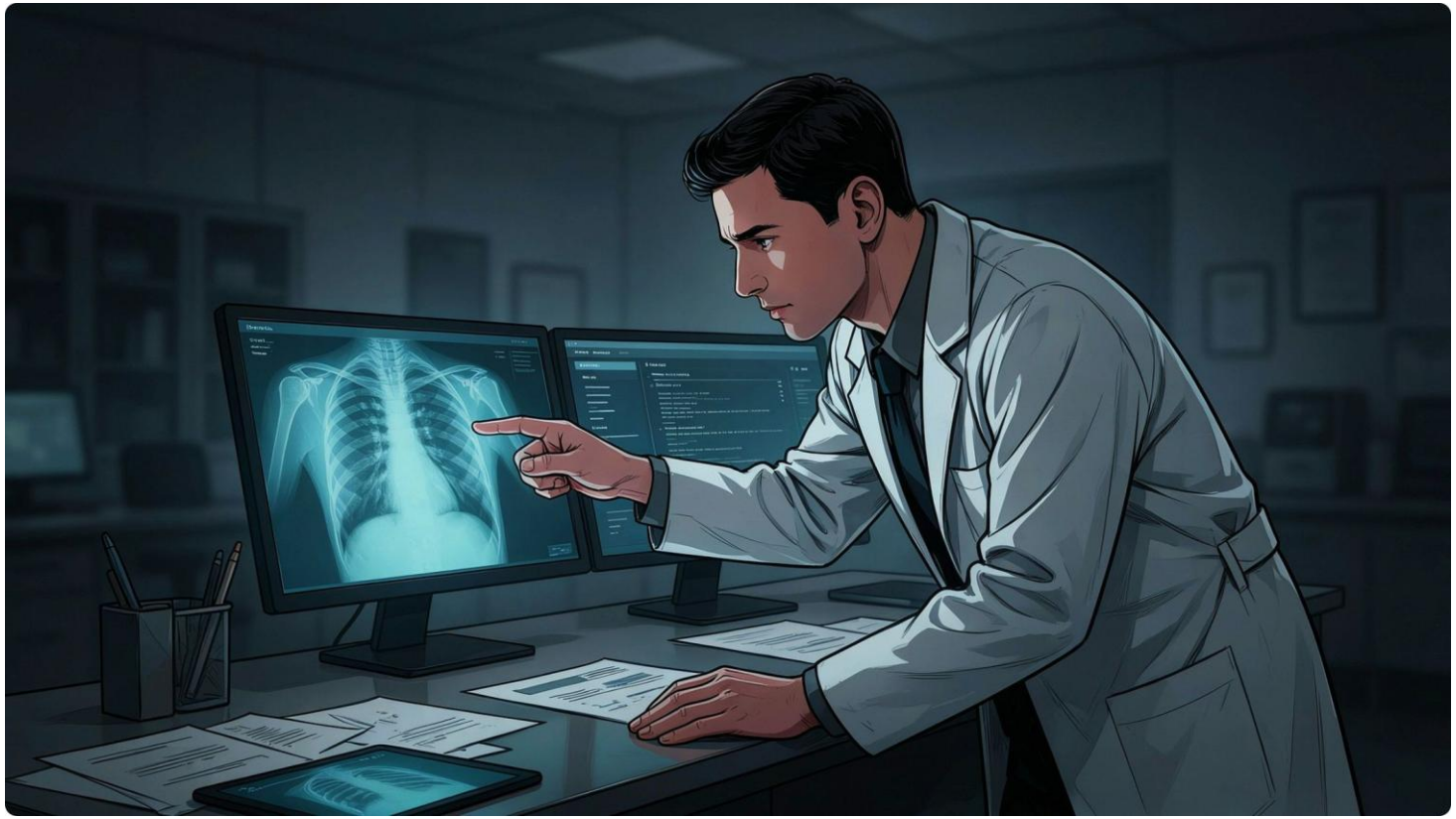
P

Physical & Embodied AI

T

Trustworthy & Sovereign AI

Multimodal Foundation Models



What They Are

Large models pre-trained on broad data, adapted to many tasks. They accept **text, images, audio, video, sensor data, documents and code** — all converted into a shared representation space.

Key Uses

- Reading medical images with clinical reports
- Voice-based learning in Indian languages
- Robots combining vision, speech and touch
- Accessibility tools

Exam tip: A text-only model cannot compare a figure with a table — a multimodal model can.

Reasoning-Oriented Models

Traditional vs. Reasoning Models

Traditional: Recognise and predict in one fast pass — strong at perception, weak on multi-step problems.

Reasoning: Break down → analyse → verify → solve. They write chain-of-thought, check partial results and call tools like calculators.

⚠ Caution: Reasoning does NOT eliminate hallucination. Verification and human judgement remain essential.

Worked Example

1,200 records; 15% duplicates; 20% of remainder fail screening.

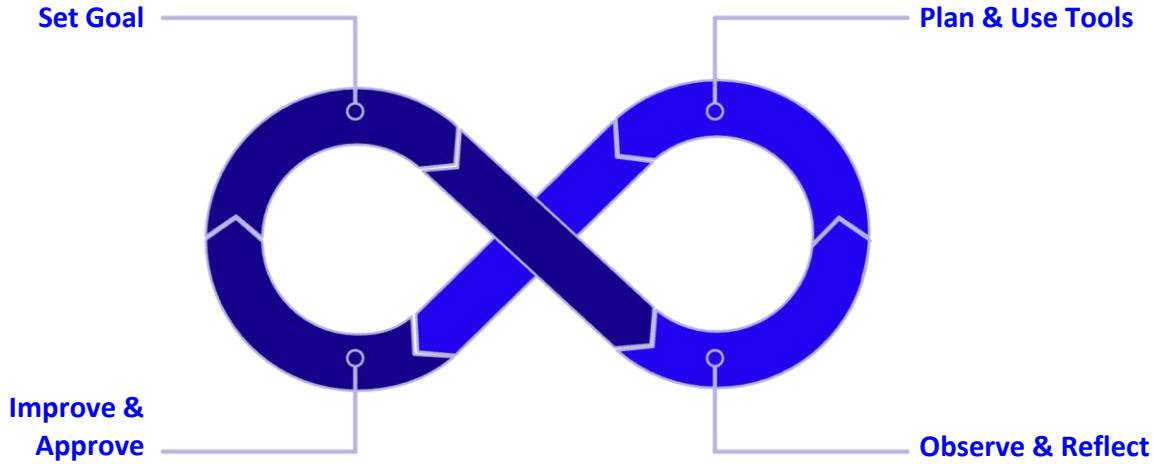
- Duplicates = 180; Unique = 1,020
- Fail screening = 204; Remaining = 816
- Check: 816 + 204 + 180 = 1,200 ✓

A one-pass model might wrongly compute 1,200 × 0.65 = 780.



Agentic AI

An AI agent pursues a **goal through many coordinated actions**, using tools, keeping task state and correcting itself — unlike a chatbot that responds one prompt at a time.



A wrong agent action (deleting a file, sending an email) is more serious than a wrong chatbot answer.

Multi-Agent Systems



Planner

Breaks the goal into tasks



Researcher

Gathers facts and sources



Coder

Implements and runs code



Reviewer

Critiques and tests output



Safety Monitor

Flags risky actions

> "Specialisation improves capability, but coordination determines reliability."

Error propagation: A hallucinated fact from the researcher agent is built upon by later agents, contaminating the whole result.

Small & Efficient Models

Large Cloud vs. Small Edge

Cloud	Edge
Pay-per-use	Low, predictable cost
Network latency	Millisecond response
Data leaves device	Data stays on device
Needs connectivity	Works offline

Key Techniques

- **Quantisation:** 7B model at 16-bit ≈ 14 GB; at 4-bit ≈ 3.5 GB
- **Distillation:** Small "student" imitates large "teacher"
- **Pruning:** Removes low-contribution weights
- **LoRA:** Trains 65,536 parameters instead of 16.8 million for one layer (~0.4%)

Cost of GPT-3.5-level performance fell over 280-fold between late 2022 and late 2024.

AI for Science & Engineering

Drug Discovery

Molecule and drug repurposing candidates

Materials Science

Batteries, catalysts, semiconductors

Healthcare Imaging

Diagnosis and clinical support

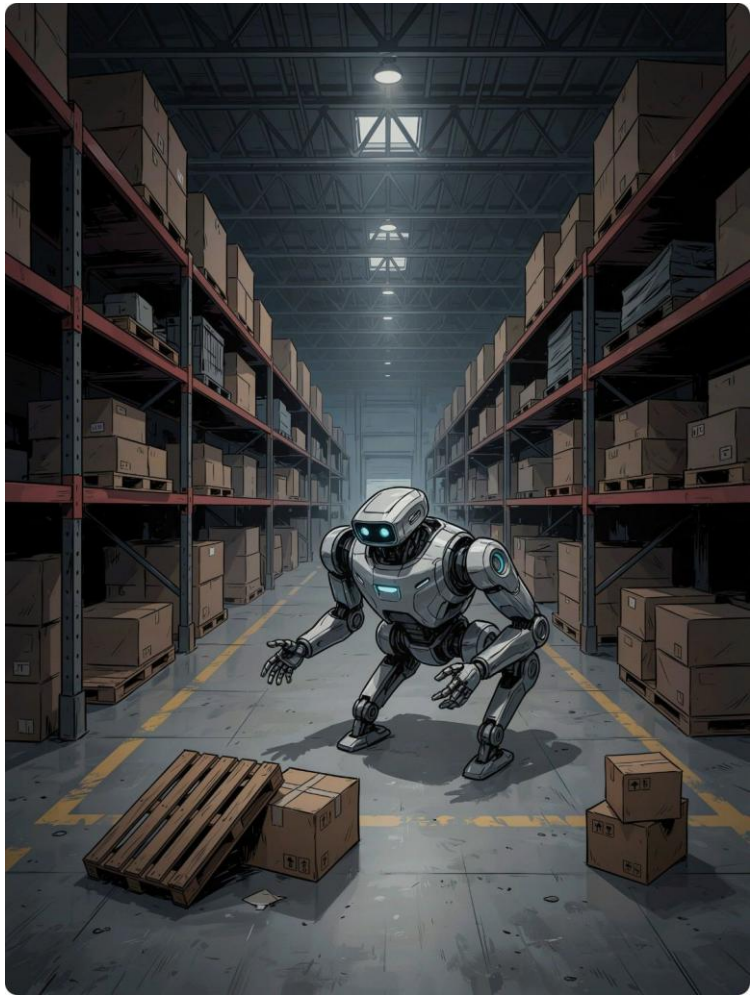
Climate & Agriculture

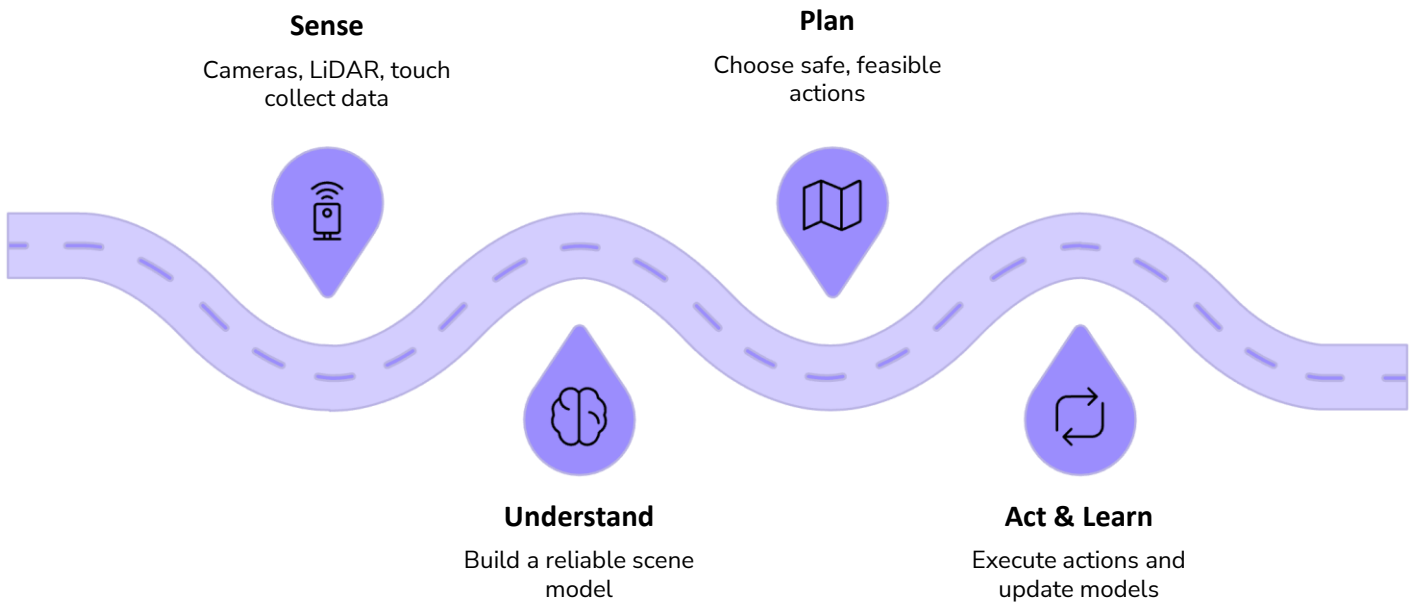
Forecasting and earth observation

AlphaFold (protein structure prediction) won the 2024 Nobel Prize in Chemistry. Google's AI co-scientist proposed liver fibrosis drug candidates validated in human organoids.



Physical & Embodied AI





"A wrong chatbot answer is inconvenient. A wrong robotic action can be dangerous." Safety testing is central.

Progress is uneven: AI agents improved sharply on computer-use benchmarks, but robots still fail most real household tasks.

AI in Software & Security

Software Engineering

- Code generation & debugging
- Test generation
- Documentation
- Legacy system understanding

Cybersecurity

- Threat & anomaly detection
- Phishing & fraud detection
- Incident response
- Vulnerability assessment

Dual-Use Warning

AI strengthens defenders **and** attackers. Generated code must be tested for correctness, security, bias and licensing.

Trustworthy & Sovereign AI

> "Capability without trust will not be adopted — and should not be."

Fairness

No group left behind by data or design

Transparency

Explainable decisions, documentation

Privacy

Data minimisation, consent

Security

Resistance to attack and misuse

Robustness

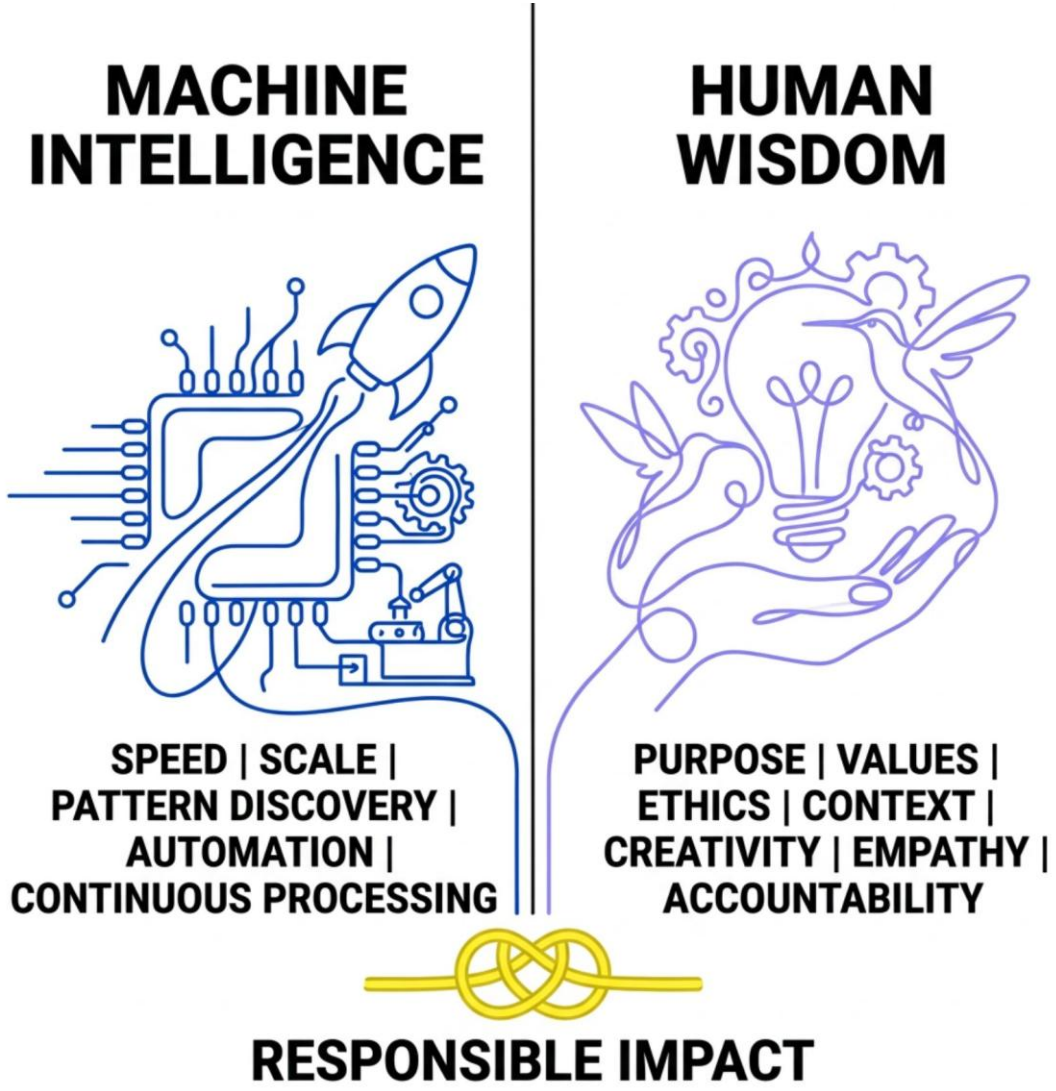
Reliable under noise, shift and stress

Accountability

Ownership, audit trails, redress

Sovereign AI: A nation's or institution's ability to build, host and govern AI on its own terms — its own data, languages, infrastructure and laws.

The Human–AI Partnership



AI Capability × Human Judgement = Responsible Impact

If either is zero, the result is zero.

Roadmap: AI User → AI Creator

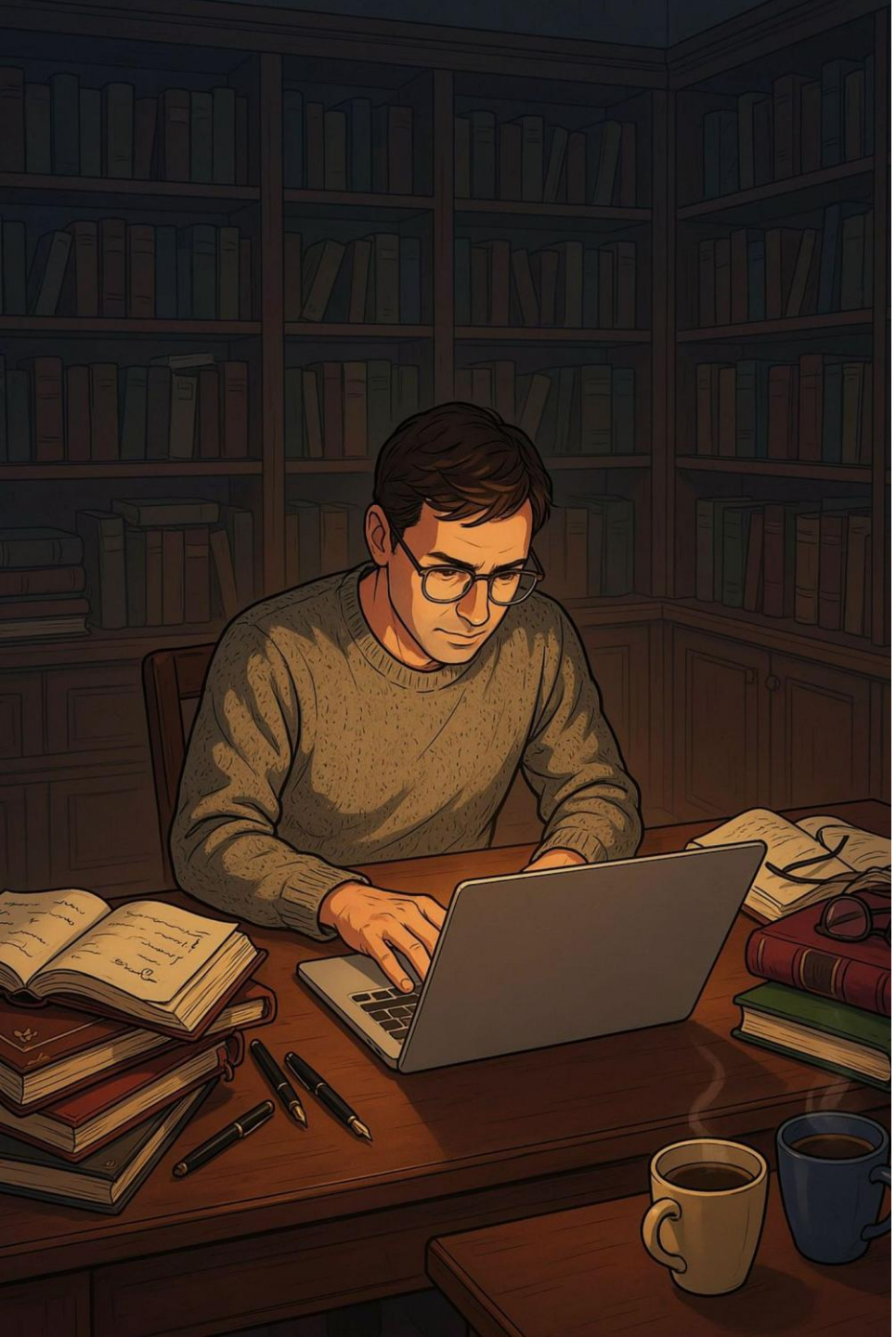
1
2
3
4

1 Foundations
Maths, algorithms, programming, databases

2 AI Core
ML, DL, NLP, vision, optimisation

3 Modern AI
Transformers, generative AI, agents, retrieval

4 Responsible Deployment
MLOps, evaluation, security, governance



PART A · SECTION 2

Foundations of AI-Assisted Research

Why AI in Research — Three Levels

1 **Access**

More literature and data than anyone can read — over 2 million papers published per year

2 **Assistance**

AI can search, summarise, code, analyse and write across the research workflow

3 **Accountability**

Claims still need human verification — responsibility does not transfer to AI

> AI assists; evidence and researcher judgement decide.

Evolution of Research Practice

1. **Traditional**

   Physical libraries, card catalogues, manual notes

2. **Digital**

   Online databases, reference managers, statistical software

3. **Data-Driven**

   Big data, simulation, ML, cloud/HPC

4. **AI-Assisted**

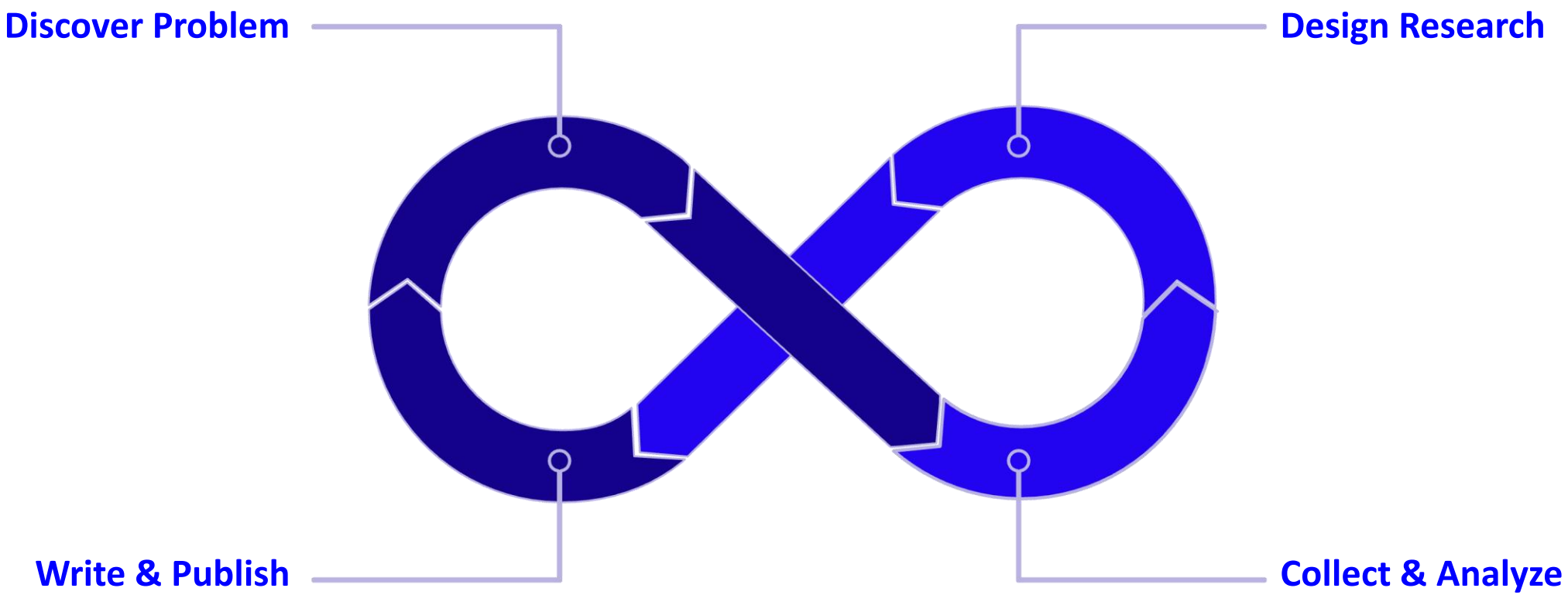
   LLM-based conversational assistance across the workflow

5. **Agentic**

   Multi-step AI workflows — emerging, not yet standard

The bottleneck is shifting from **finding information** to **evaluating information**.

The Research Lifecycle



It is **iterative, not linear**. AI can assist at every stage, but human verification is required at every stage. **The ability to assist does not imply the authority to decide.**

Human–AI Collaboration in Research

AI's Potential Strengths

- Speed, scale, repetitive processing
- Summarisation and pattern identification
- Generating alternatives and formatting
- Coding help and language transformation

Human Responsibilities (Always)

- Choosing meaningful problems
- Judging scientific validity
- Distinguishing causal from correlational claims
- Ethics, interpretation and final claims

The researcher is **epistemically responsible** — "the AI said so" is never a justification.

Three Levels of AI Use

AI as a Tool

Formatting, rewriting, code help — lowest autonomy

AI as an Assistant

Search, summarisation, comparison, extraction — medium autonomy

AI as a Collaborative System

Multi-step reasoning, planning, workflows — highest autonomy

⚠ As autonomy increases, verification must become **more** systematic, not less.

AI output is an intermediate research object — a plausible but unverified candidate that becomes evidence only after the researcher evaluates it against primary sources.

AI-Assisted vs. AI-Generated Research

AI-Assisted ✅

- Human defines the problem
- AI accelerates tasks
- Evidence is verified
- Researcher interprets and owns decisions

AI-Generated ❌

- AI drives conclusions
- Sources go unchecked
- Reasoning is opaque
- Hallucination risk rises
- Accountability becomes unclear

Human expertise × AI assistance × verification = responsible research

Vibe Coding



What It Is

Conversational, rapid, AI-driven creation and modification of software through iterative natural-language specification.

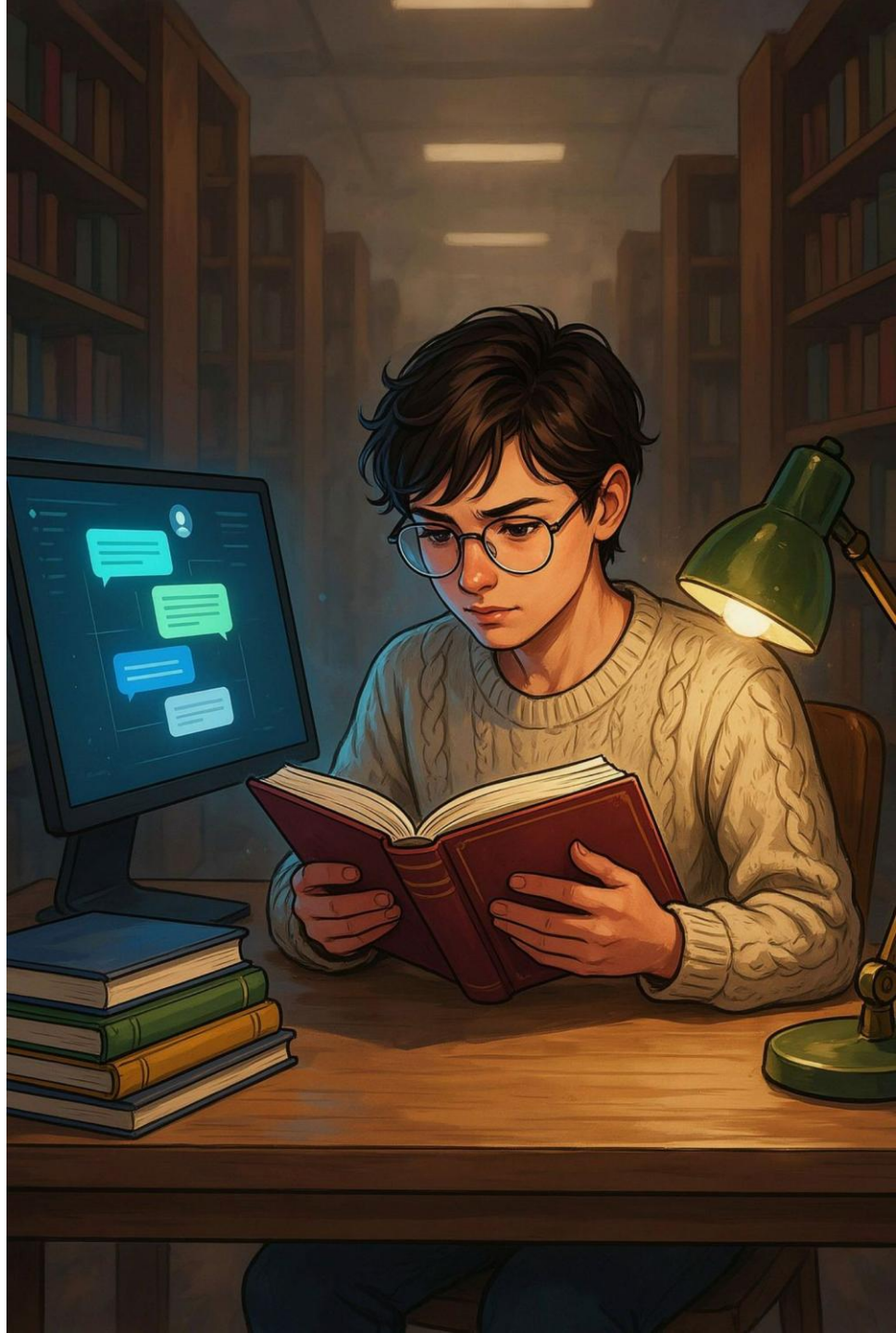
Good Practice

- Review and test all generated code
- Verify dependencies
- Watch for technical debt
- Ensure reproducibility

Beware: silent errors, security flaws, unverified dependencies. Research code must survive testing and reproduction.

PART A · SECTION 3

AI Literacy



What AI Literacy Means

Understand

What the tool does and what model/data underlie it

Use Appropriately

Apply the right tool for the right task

Evaluate

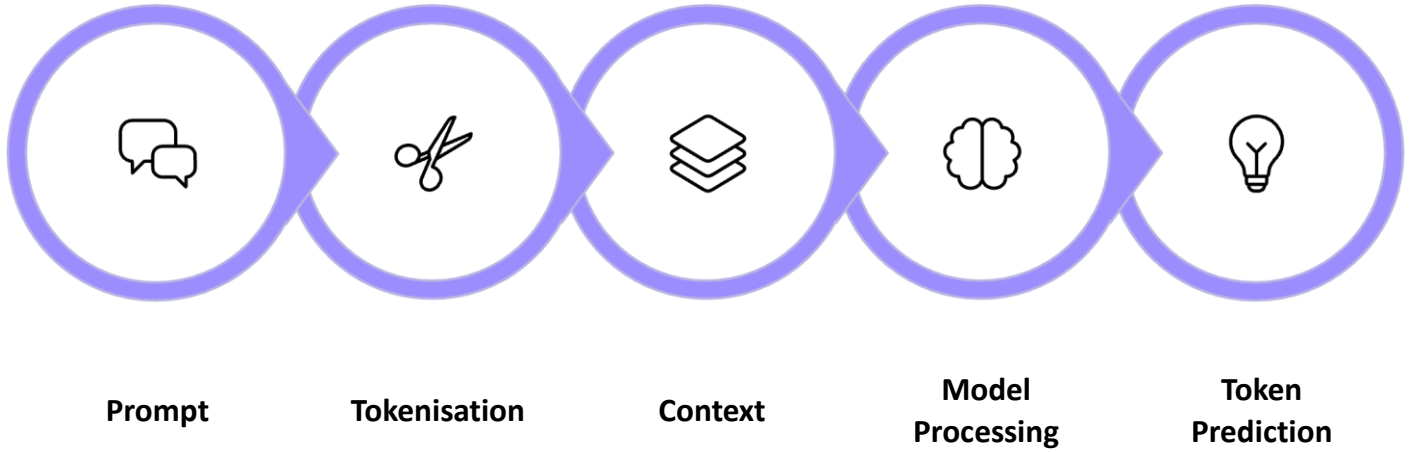
Assess reliability and bias in outputs

Verify

Check against primary sources — always

Using AI is NOT the same as being AI literate.

How LLMs Work



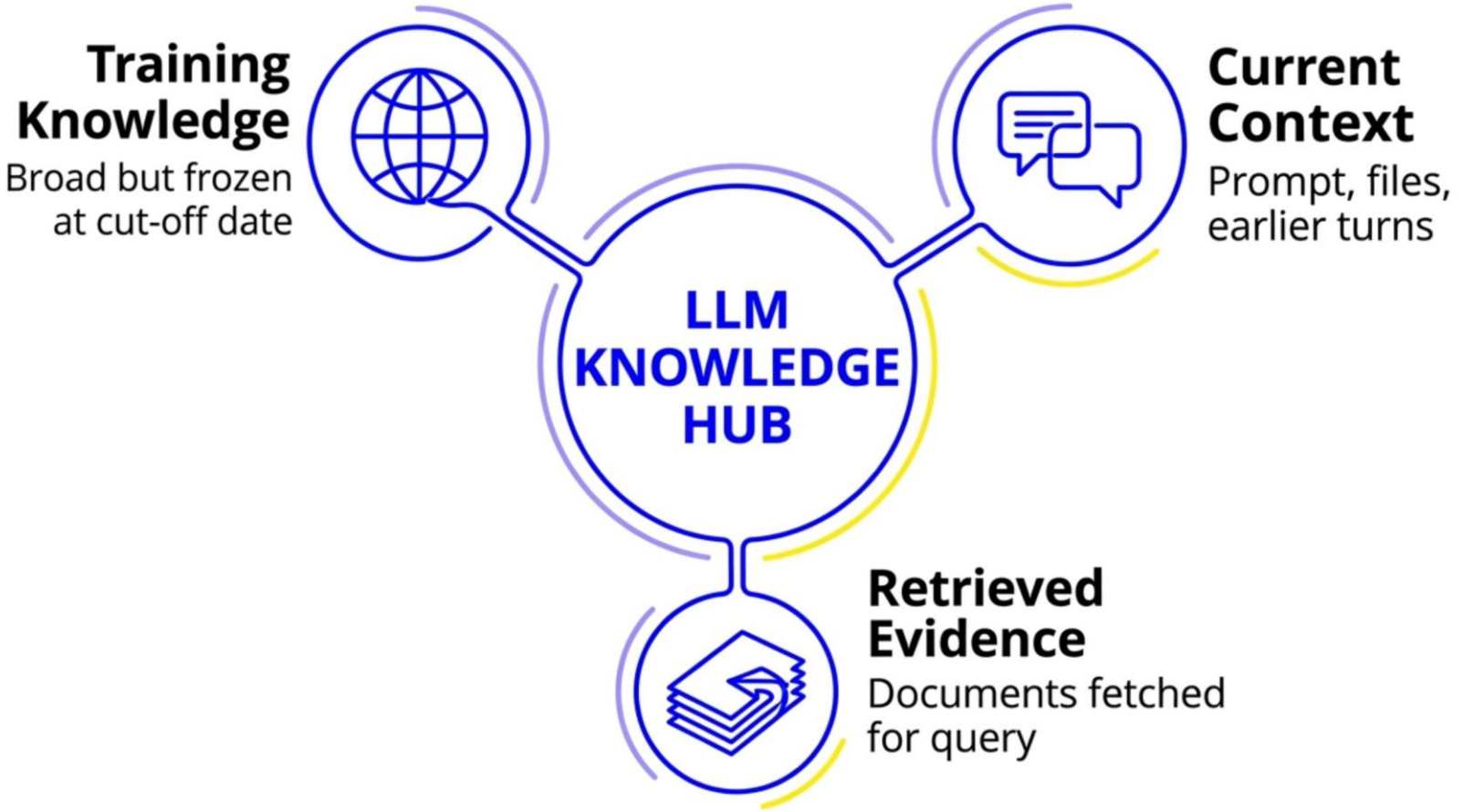
Key Facts

- Trained on huge text collections to predict the next token
- Generates from learned patterns — NOT a database lookup
- Fluent output does NOT guarantee factual correctness

Tokens

A token is a small unit of text (often a word-piece). In English, 1 token ≈ 0.75 words. A 6,000-word paper ≈ 8,000 tokens. Context limits and pricing are measured in tokens.

Context Window & Knowledge Sources



A paper published after the cut-off and not supplied in context cannot be known — though the model may still guess (and be wrong).

Grounding & RAG

What is Grounding?

Tying an answer to specific, traceable sources so each claim can be checked. An **ungrounded** response comes from internal patterns; a **grounded** response is built from retrieved sources.

RAG Process

01

Chunk & Embed

Documents split and stored as vectors

02

Retrieve

Question embedded; similar chunks fetched

03

Generate

LLM answers citing retrieved chunks

04

Verify

Human checks cited passages

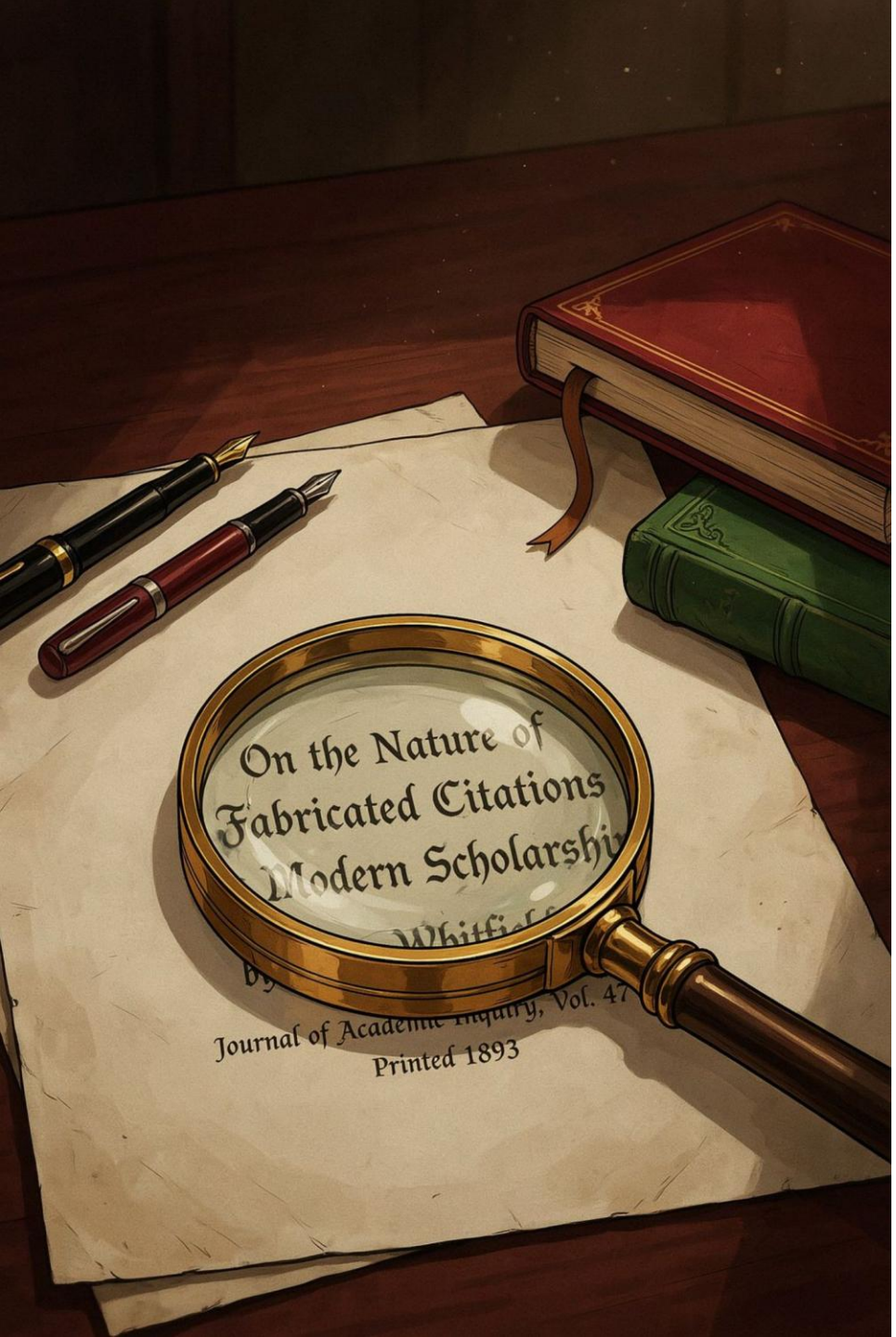
RAG Limits

- Can retrieve wrong chunks
- Can misread the right chunks
- Can ignore retrieved sources

⚠ RAG makes evidence **available** for verification — it does NOT guarantee correctness.

GraphRAG

Retrieves connected entities and relations from a knowledge graph (e.g., drug → target → disease), helping with multi-hop questions.



Hallucination — The #1 Risk

Wrong Fact

Incorrect information stated confidently

Invented Citation

Author, title, journal, year and DOI — all fake but professional-looking

Wrong Interpretation

Real paper, real citation — but misrepresented claim

An audit of NeurIPS 2025 papers found ~100 hallucinated citations in ~50 papers that had passed review. Major conferences now treat hallucinated references as grounds for desk rejection.

How to Verify a Citation

1 **Does the paper exist?**

Paste the DOI into doi.org — does it resolve?

2 **Is it findable?**

Search title on Google Scholar or Semantic Scholar

3 **Is the journal indexed?**

Confirm in Scopus or Web of Science

4 **Does it support the claim?**

Open the paper and find the exact sentence — check context

If any step fails, do not cite it.

Other Ways AI Misleads

Missing Context

Accurate in isolation but misleading — e.g., "95% accuracy" when 95% of cases are negative

Automation Bias

Trusting machine output without verification because it sounds confident

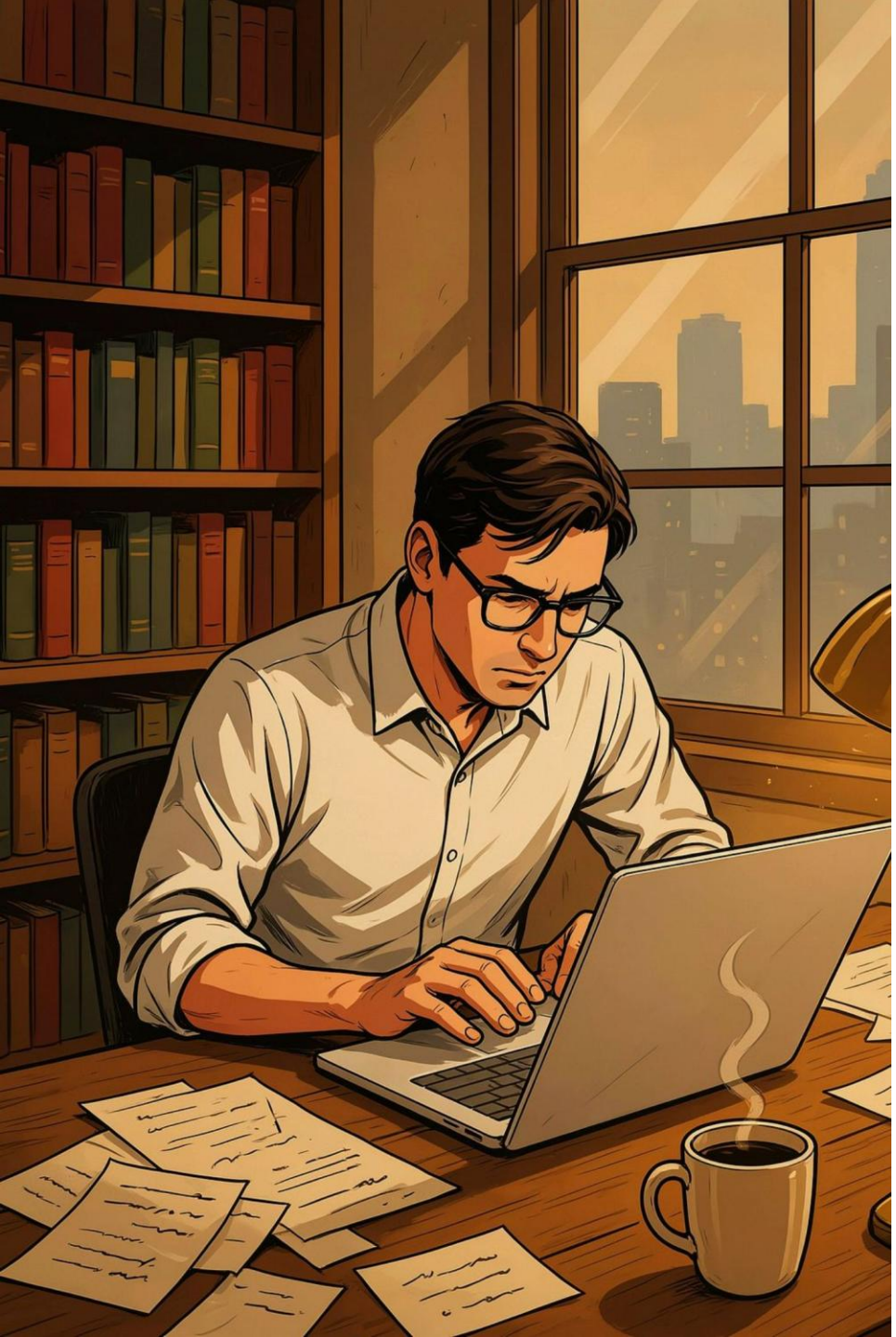
Overconfidence

Uncertainty hidden behind fluent language; AI rarely says "I don't know"

Further Risks

Bias, outdated information, privacy leaks, plagiarism, poor reproducibility, over-dependence, loss of critical thinking

Golden rule: Trust, but verify. The researcher's new skill: find → question → verify → synthesise → judge.



PART A · SECTION 4

Prompt Engineering for Research

Prompts as Research Instructions

Weak Prompt ❌

"Find papers on concept drift"

No domain, time range, source type, output format or verification requirement — AI returns whatever seems plausible, possibly fabricated.

Strong Prompt ✅

"Identify peer-reviewed studies on concept drift detection in healthcare data streams published between 2021 and 2026. Provide title, authors, venue, year and DOI. Do not invent references."

> "A powerful AI tool with a poor research question is still a poor research process."

The R–C–S–E–O–V Framework

R — Role

Who the AI acts as and at what depth. Clarifies expertise — not a magic phrase.

C — Context

Domain, problem, objective, prior knowledge. **The single most powerful part.**

S — Scope

Years, domain, study type, what to exclude. A methodological decision.

E — Evidence

Acceptable sources, required details. "Do not invent references."

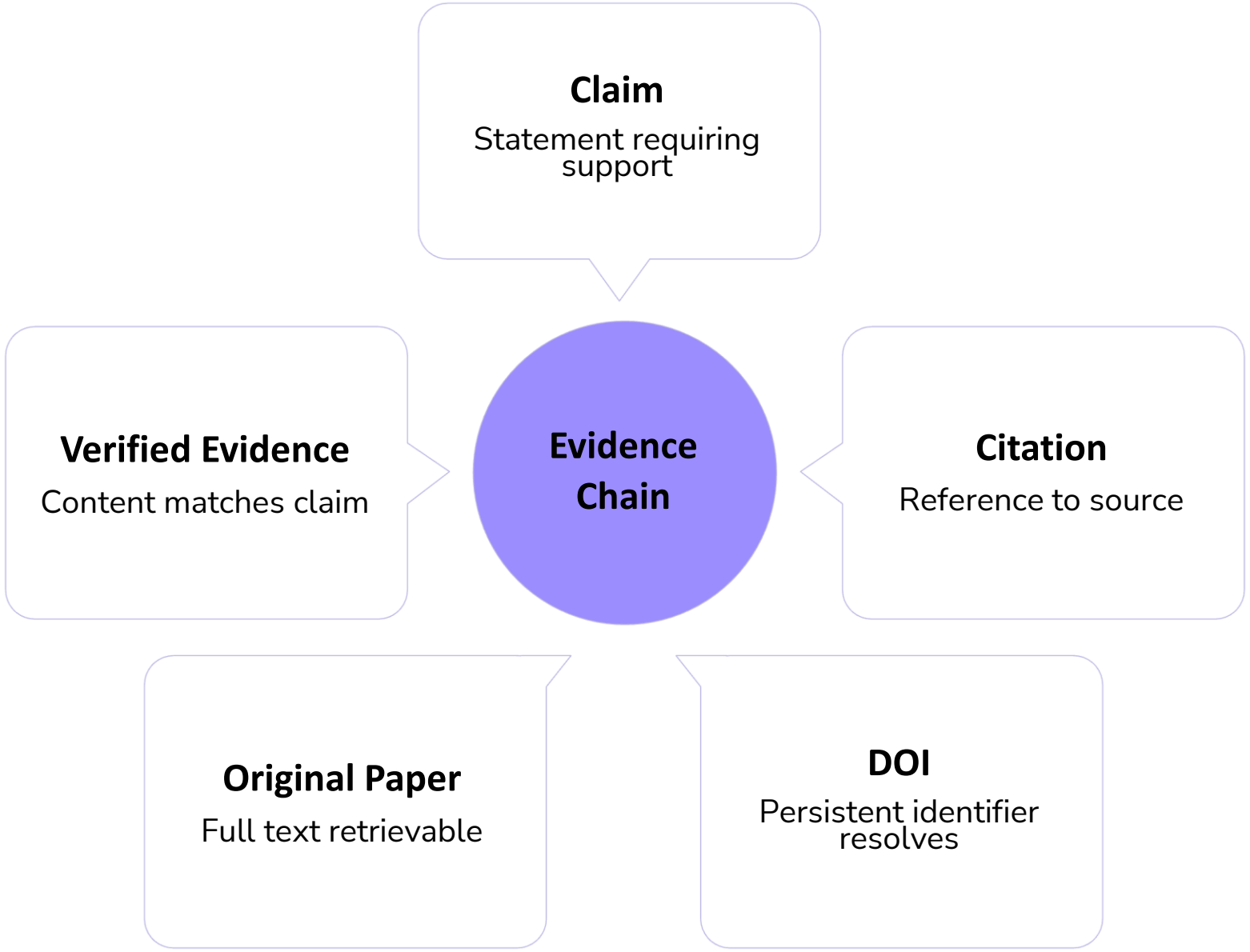
O — Output

Structure: table of paper, method, dataset, finding, limitation.

V — Verification

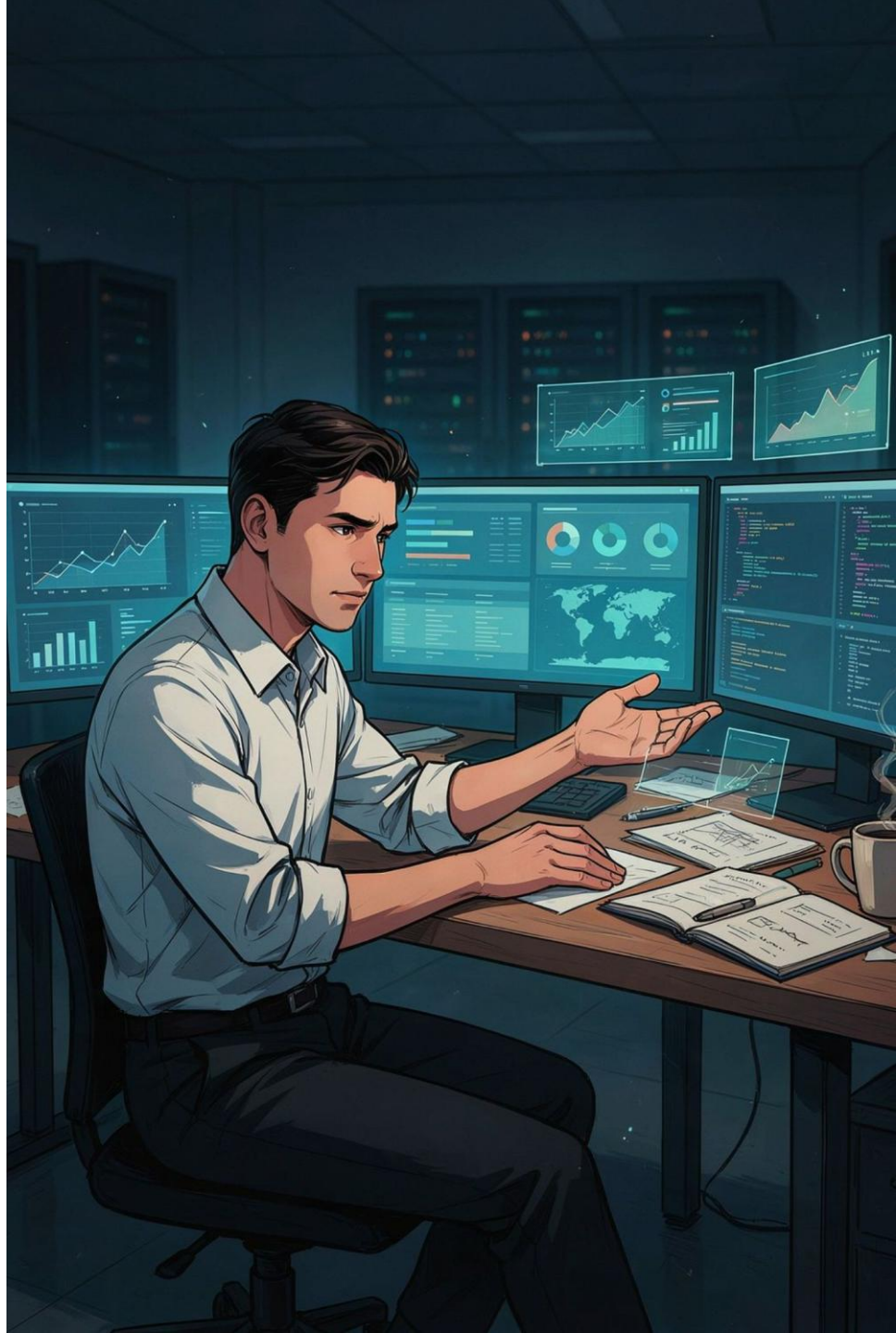
"Mark anything that cannot be independently verified." Design this in — not at the end.

The Evidence Chain



PART A · SECTION 5

Evaluating AI Research Tools



Three Questions for Any AI Tool

1

Source Base

Where does the information come from? LLM knowledge, web search, scholarly database, citation index or knowledge graph? Is the corpus documented?

2

Model Capability

What can it actually do? Search, retrieve, summarise, generate, analyse citations, map knowledge or detect trends?

3

Reliability

Can results be independently verified? Evaluate the research infrastructure, not only the interface.

Source Types Compared

Source Type	Strengths	Weaknesses
LLM Internal Knowledge	Broad coverage	Training cut-off; high hallucination risk
Web Search	Broad, grey literature	Unfiltered
Scholarly Databases	Curated, precise	May have gaps
Citation Indexes	Impact & network analysis	Subscription required
Knowledge Graphs	Programmatic, relational	Requires technical skill

Scholarly databases and knowledge graphs are **preferred** for research.

Tool Evaluation Scorecard

Six Dimensions (1–5 each, max 30)

- Source quality
- Coverage
- Citation traceability
- Freshness (LLM cut-offs miss last 12–24 months)
- Transparency
- Reproducibility

Score based on documented evidence, not marketing.

Example Scores

~9

General Chatbot

Poor traceability & reproducibility

~25

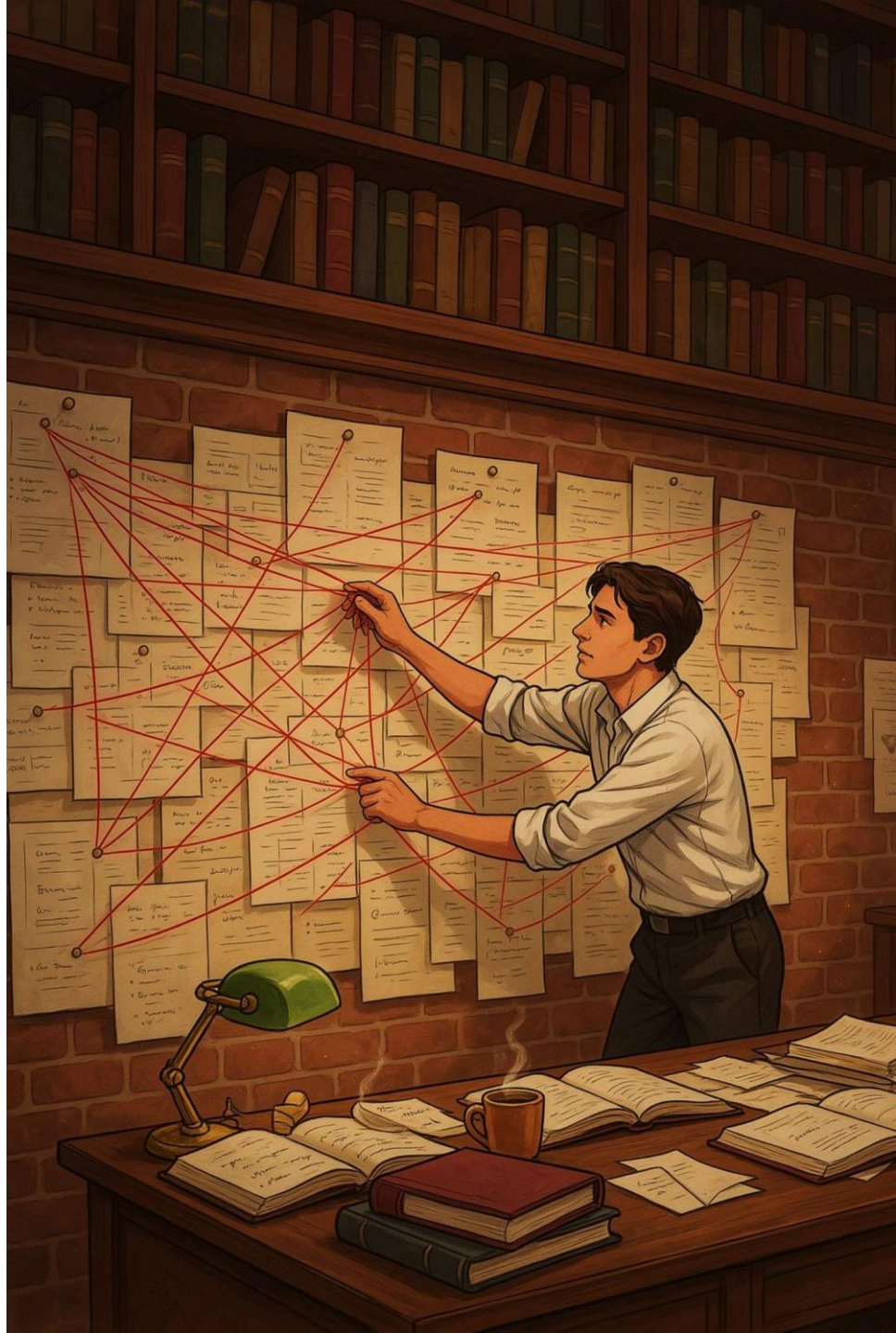
Scholarly Search Engine

Strong on all dimensions

Choose tools by research task, not popularity.

PART A · SECTION 6

Literature Discovery



Search vs. Discovery

Search

Keyword query → matching results

Fast but limited to what the query names. Misses synonyms and different terminology.

Discovery

Seed paper → references (backward) → citations (forward) → authors → clusters → emerging themes

Finds foundational papers, emerging work not yet cited, and influential authors on adjacent problems.

> Search retrieves; discovery explores relationships.

What AI Can and Cannot Do in Literature Review

AI Can Accelerate ✅

- Search and discovery
- Screening
- Initial comparison
- Summarisation

AI Cannot Replace ❌

- Critical evaluation
- Evidence synthesis
- Gap identification
- Formulating the research question

⊗ Collapsing everything into one "generate literature review" prompt produces a simulacrum of a review, not a scholarly one.

Seed Papers & Citation Networks

What is a Seed Paper?

A single trusted, highly relevant paper used as the entry point into a field. Identify **2–5 seed papers** before using discovery tools.

From Seeds You Reach:

- **Prior works** — papers the seed cites (where the idea came from)
- **Derivative works** — papers citing the seed (where the idea went)
- Similar papers, authors, institutions and topics

Citation Count ≠ Quality

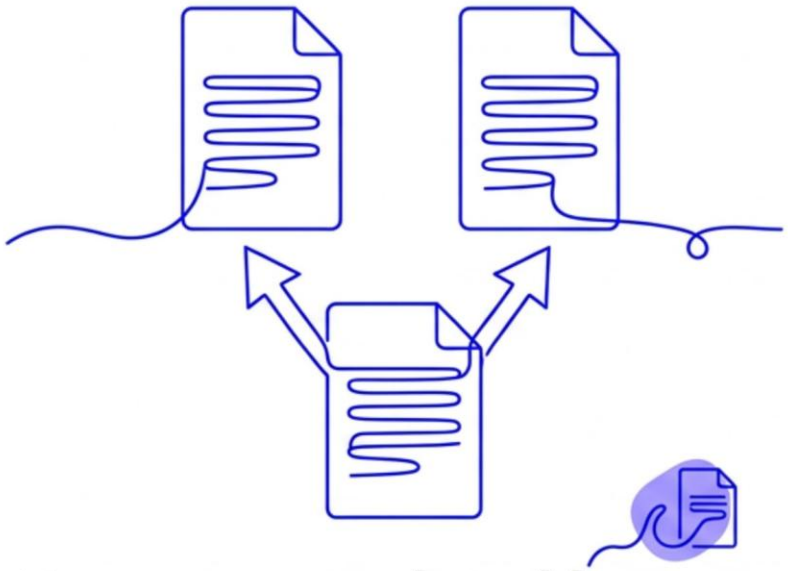
A 2015 paper with 5,000 citations vs. a 2025 paper with 25 citations — raw counts are incomparable.

- Older papers accumulate more citations
- Large fields generate more citations
- A highly cited paper may address a different problem
- Citations do not measure rigour

Citation count signals **influence**, not relevance or quality.

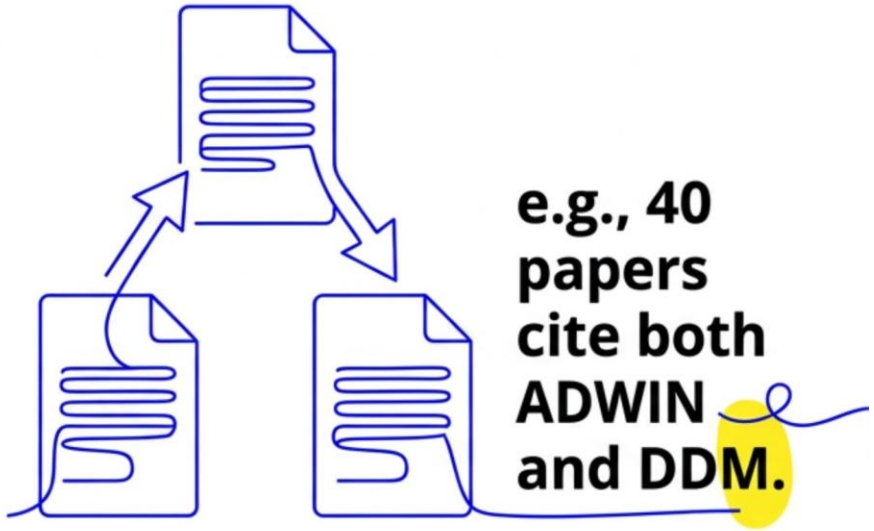
Co-citation & Bibliographic Coupling

BIBLIOGRAPHIC COUPLING



Two papers related because they **cite the same** earlier works (backward-looking).

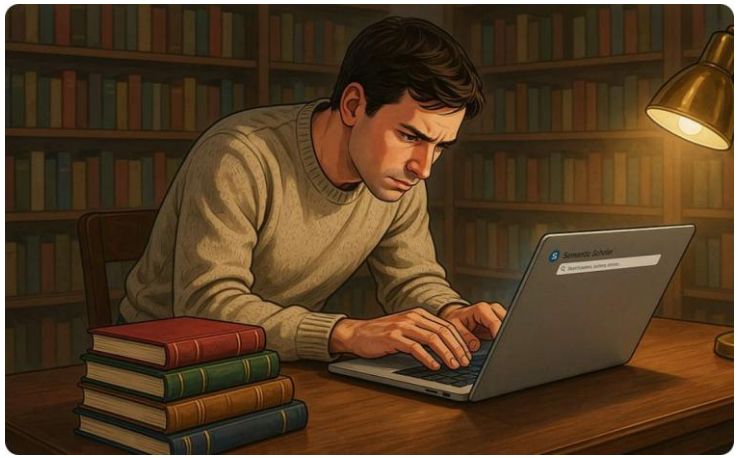
CO-CITATION



Two papers related because later papers **cite both** (forward-looking).

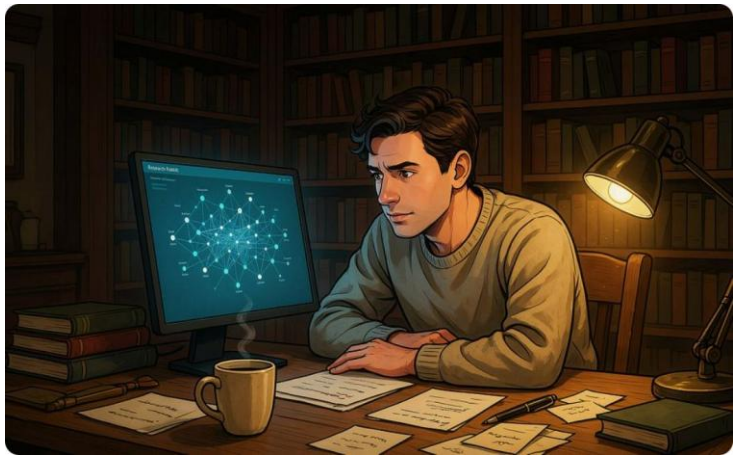
These relationships power similarity maps in tools like Connected Papers.

Discovery Tools — Part 1



Semantic Scholar

Free AI-powered scholarly search from Allen Institute for AI. Indexes 200M+ papers. Strategy: search → anchor paper → explore references → citations → authors → refine.



Research Rabbit

Visualises citation relationships around a seed collection. Best for expanding a literature network over time. Acquired by Litmaps in 2025.



Connected Papers

Builds a similarity graph from one seed using co-citation and bibliographic coupling. Best for quickly understanding field structure.

Discovery Tools — Part 2

OpenAlex

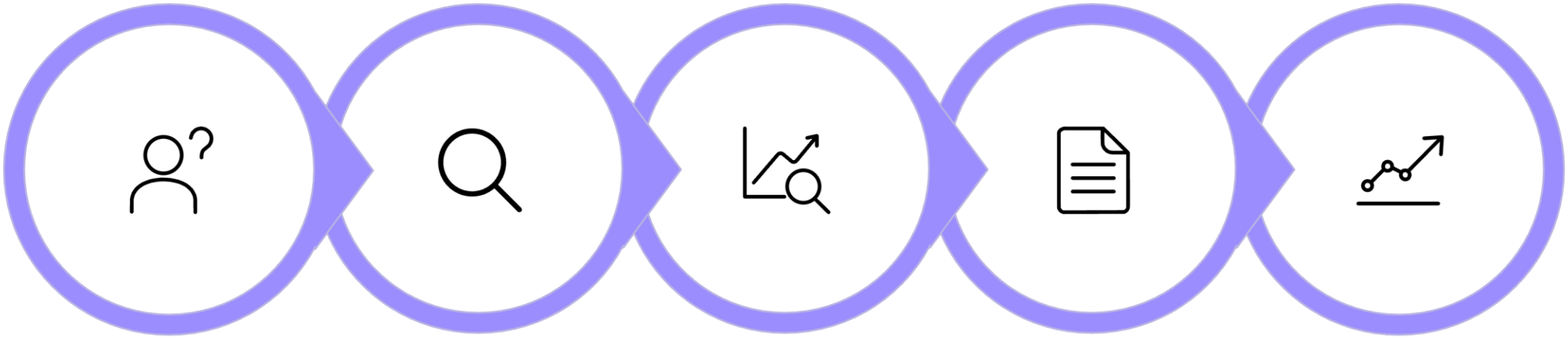
Fully open scholarly knowledge graph of ~477 million works. Used for publication trends, institutional comparisons, collaboration networks and topic emergence — literature analysed as data.

Litmaps

Plots papers on a publication timeline with citation links and alerts you to new relevant papers. Ideal for **longitudinal monitoring** across a multi-year PhD. Literature review is continuous, not one-time.

There is no single best tool — only a best tool for a specific task. Tool orchestration matters more than tool collection.

Integrated Discovery Workflow



Define **Discover** **Expand** **Map** **Analyse**

The researcher begins and ends the process; AI accelerates the middle. Always verify evidence in primary sources before synthesising.

Common Mistakes & Good Rules

Common Mistakes ❌

- Asking AI for "20 references"
- Copying AI-written reviews
- Trusting citations without checking
- Searching only through chatbots
- Equating citations with quality
- Using one tool for everything
- Letting AI declare the research gap

Good Rules ✅

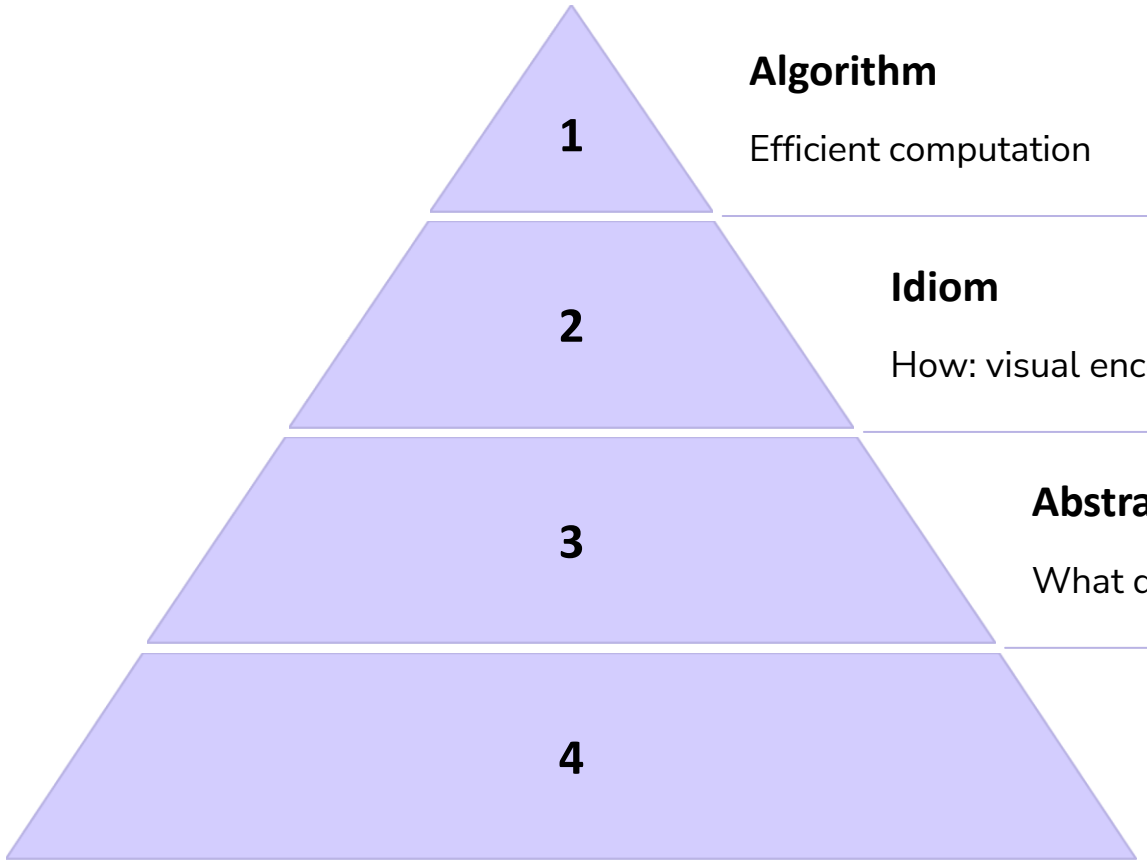
- Search before synthesis
- Verify important citations
- Prefer primary scholarly sources
- Separate evidence from AI interpretation
- Use multiple discovery strategies
- Keep humans in the decision loop
- Document meaningful AI assistance



PART A · SECTION 7

Data Visualization Principles

Munzner's Four-Level Nested Model



Algorithm

Efficient computation

Idiom

How: visual encoding & interaction

Abstraction

What data & why the task

Domain Situation

Who the users are & what they need

Levels are nested: if the domain need is misunderstood, a perfect chart still fails.

Data & Task Abstraction

Attribute Types

- **Categorical:** No order (shapes, fruit types)
- **Ordinal:** Ranked but no meaningful differences (S/M/L)
- **Quantitative:** Meaningful arithmetic (temperature, length)

Ordering Direction

- Sequential, Diverging (around a midpoint), Cyclic

Task Types (Search)

- **Lookup:** Target and location both known
- **Browse:** Location known, target not
- **Locate:** Target known, location not
- **Explore:** Neither known

Task Format

Written as {action, target} pairs: e.g., "discover distribution" or "locate outliers"

Channels: Expressiveness & Effectiveness

Magnitude Channels (Best → Worst)

01

Position on common scale

02

Position on unaligned scales

03

Length

04

Angle / Area / Depth

05

Colour luminance & saturation

Key Definitions

Expressiveness: Channel matches the data type (don't encode quantities with hue).

Effectiveness: Most important data uses the most accurately perceived channels.

Bar vs. Pie

Bars use length on a common scale (top channel). Pies use angle and area (lower channels). Readers compare 43.8% vs. 43.9% far more accurately with bars.

Colour, Grouping & Design Principles

Colour Palette Selection

- **Sequential:** Magnitudes (rainfall)
- **Diverging:** Values around a midpoint (temperature anomaly)
- **Categorical:** Distinct groups (≤8 hues)
- **Cyclic:** Months, directions

⚠ Avoid relying on red vs. green alone — colour-blind safety matters.

Grouping Methods

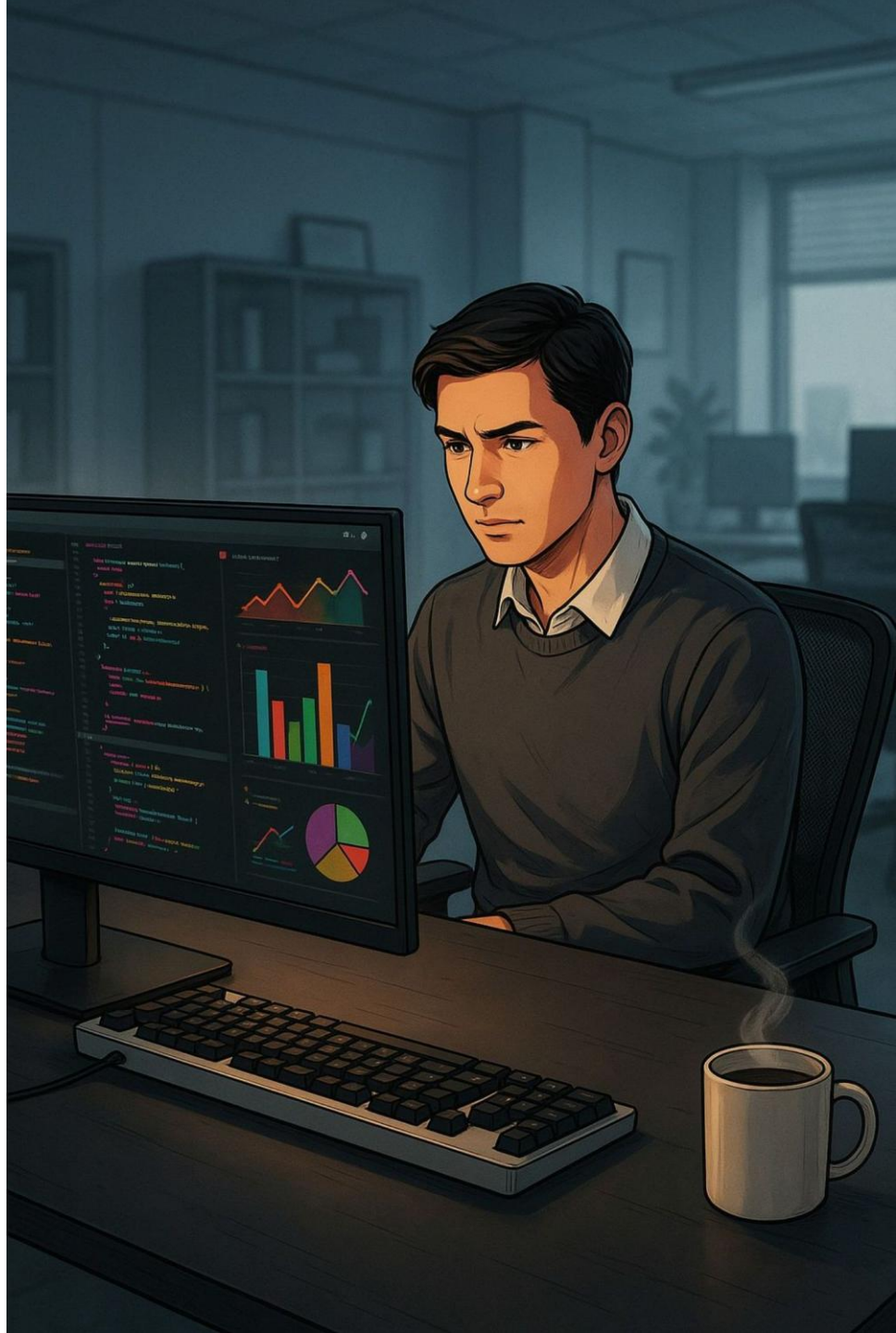
- Containment (enclosing region)
- Connection (linking lines)
- Proximity (same spatial region)
- Similarity (same colour or shape)

Four Design Principles

Proximity · Alignment · Repetition · Contrast

PART A · SECTION 8

Automated Visualization with LLMs — LIDA



Earlier AutoViz Approaches & Their Limits

Heuristic Approaches

Enumerate and rank candidate charts (DeepEye, Voyager). Tedious to maintain, limited coverage.

Task Decomposition

Specialised tools per subtask (NL4DV). Individual steps become bottlenecks (language ambiguity).

Learning-Based

Data2Vis: JSON → Vega-Lite. Tied to one grammar, needs paired training data, poor error handling, no infographics.

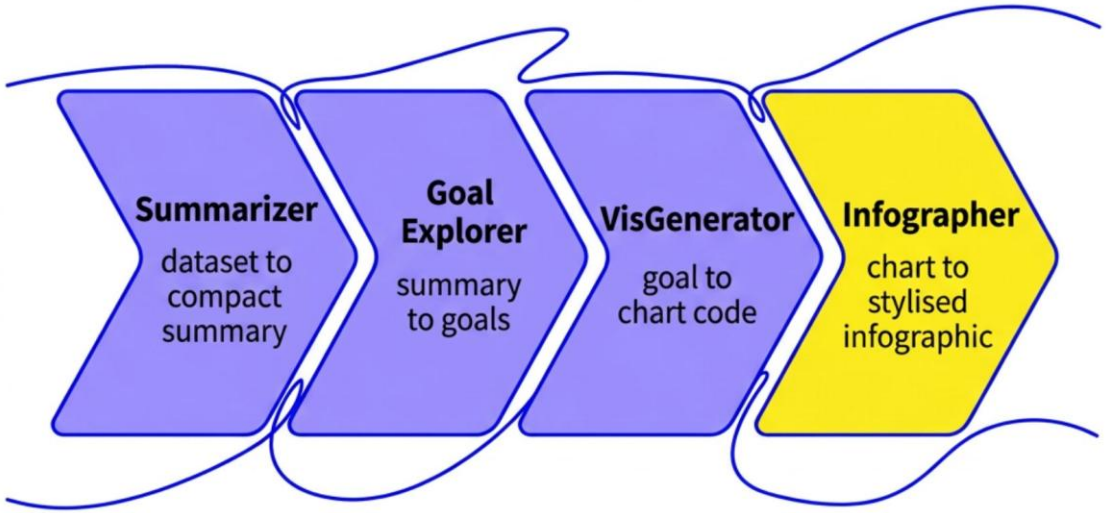
What is LIDA?

Microsoft's LLM-Based Visualization System

- Uses knowledge LLMs learned from language and code
- **Grammar-agnostic:** Matplotlib, Seaborn, Altair, ggplot
- Natural-language control
- Uses chain-of-thought, calibration and self-consistency

Core Capabilities

- Data summarisation
- Automated data exploration ("EDA for free")
- Grammar-agnostic visualisation
- Infographic generation



LIDA — The Four Modules in Detail

Summarizer

Stage 1: rules extract types, stats, sample values. Stage 2: LLM optionally adds descriptions and semantic types (e.g., a string column whose semantic type is "car model").

Goal Explorer

Generates JSON goals from the summary: {question, visualization, rationale}. Example: "What is the distribution of Miles_per_Gallon?"

VisGenerator

Three submodules: code scaffold constructor → code generator (fill-in-the-middle) → code executor. Generates, runs and filters candidates.

Infographer

Turns charts into stylised infographics using text-conditioned image diffusion. Stylisation can distort data — always check against the original chart.

Fill-in-the-Middle & Filtering

Fill-in-the-Middle Generation

The model receives a **prefix** (text before the gap) and a **suffix** (text after) and generates the middle so it fits both.

The scaffold fixes the structure → fewer structural errors. Example output:

```
alt.Chart(data).mark_point().encode(x='Miles_per_Gallon', y='Cylinders')
```

Filtering Bad Candidates

- **Compile check:** Discard code that fails to compile
- **Self-consistency:** Choose the answer most candidates agree on
- **Calibration:** Rank by estimated correctness probability

Temperature: 0 = deterministic; higher = more varied candidates. Each option trades accuracy against latency.

LIDA Evaluation Metrics

VER — Visualization Error Rate

VER = (E / T) × 100

E = charts with code errors; T = total generated.

Example: 16 of 285 charts fail → VER = 16/285 × 100 ≈ **5.6%**

⚠ A chart can compile and still be misleading — VER alone is not enough.

SEVQ — Self-Evaluated Visualization Quality

LLM-as-judge scores each chart on **six dimensions:**

- Code accuracy
- Data transformation
- Goal compliance
- Visualisation type
- Data encoding
- Aesthetics

Cheap and scalable but may share the generator's blind spots.

LIDA Benchmark Findings

3.5–9.5%

Error Rate With Summary

Across 57 Vega datasets, 5 goals each

96–99%

Error Rate Without Summary

The summary is essential grounding

75.1%

PPTC Single Instructions

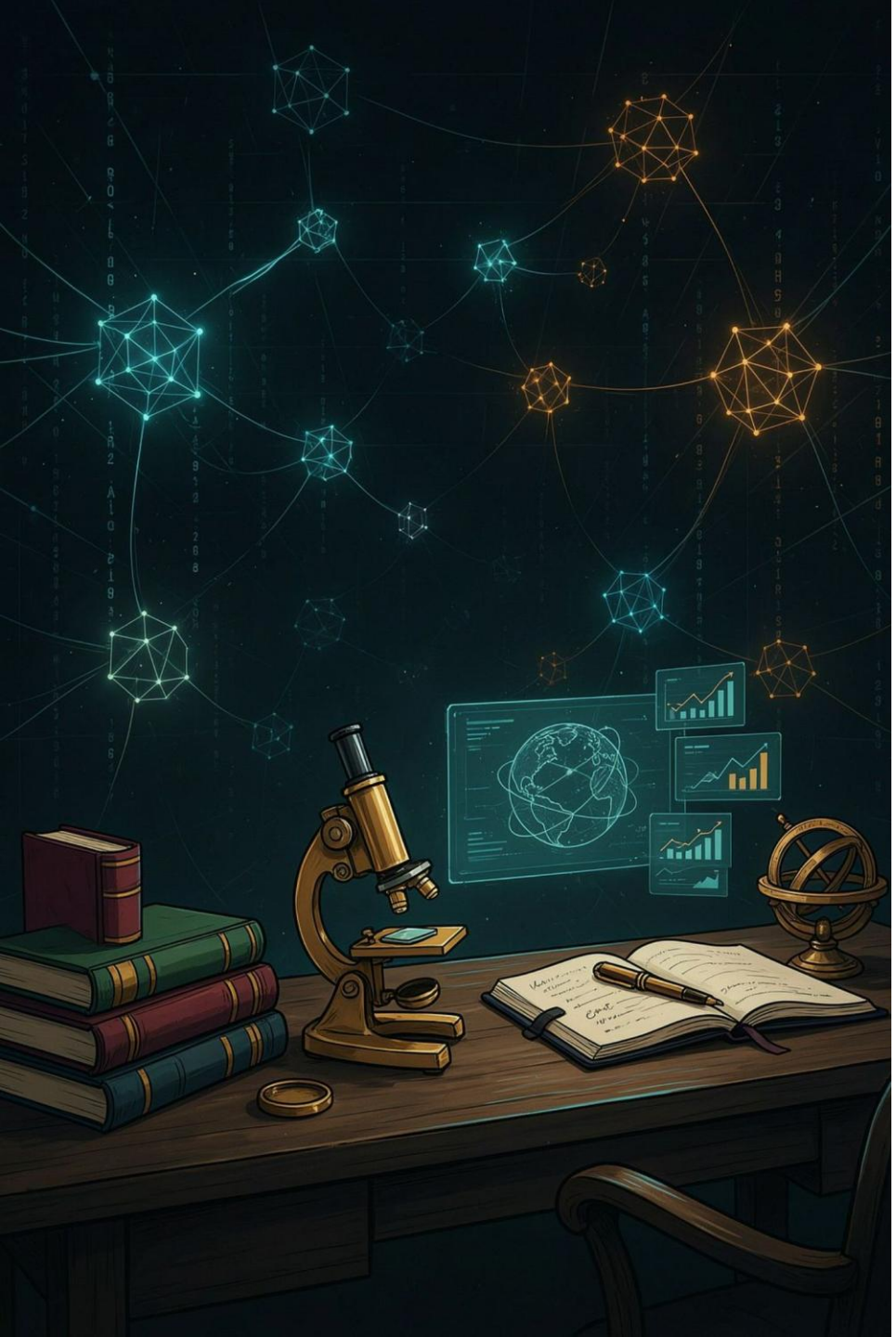
GPT-4 completion rate

~6%

PPTC Multi-Turn Sessions

Errors accumulate across steps

Install LIDA: `pip install lida`. Run: `lida ui --port=8080`. Always use a secure environment — LIDA generates and executes code.



PART A · SECTION 9

The AI Research Tool Ecosystem

Eras of AI for Science

1970s–1990s

Rule-based & expert systems (MYCIN, DENDRAL)

1990s–2012

Classical ML: SVMs, random forests, word2vec

2012–2019

Deep learning & neural NLP: BERT, BioBERT

2020–2022

Foundation models: GPT-3, AlphaFold 2

2022–2023

Instruction-tuned LLMs: ChatGPT, GPT-4, Claude

2023–2025

Agentic & multimodal AI: ReAct, AutoGen, CrewAI

2025+

AI co-scientists: hypothesis → experiment → paper

Six Categories of AI Research Tools



General-Purpose Assistants

ChatGPT, Claude, Gemini — broad reasoning, writing, coding



Literature Review & Search

Elicit, Consensus, Research Rabbit, Semantic Scholar, NotebookLM, Perplexity



Writing Assistance

Paperpal, Writefull, Jenni AI, Grammarly



Code & Data Analysis

Julius AI (quantitative), NVivo, ATLAS.ti (qualitative)



Domain-Specific Tools

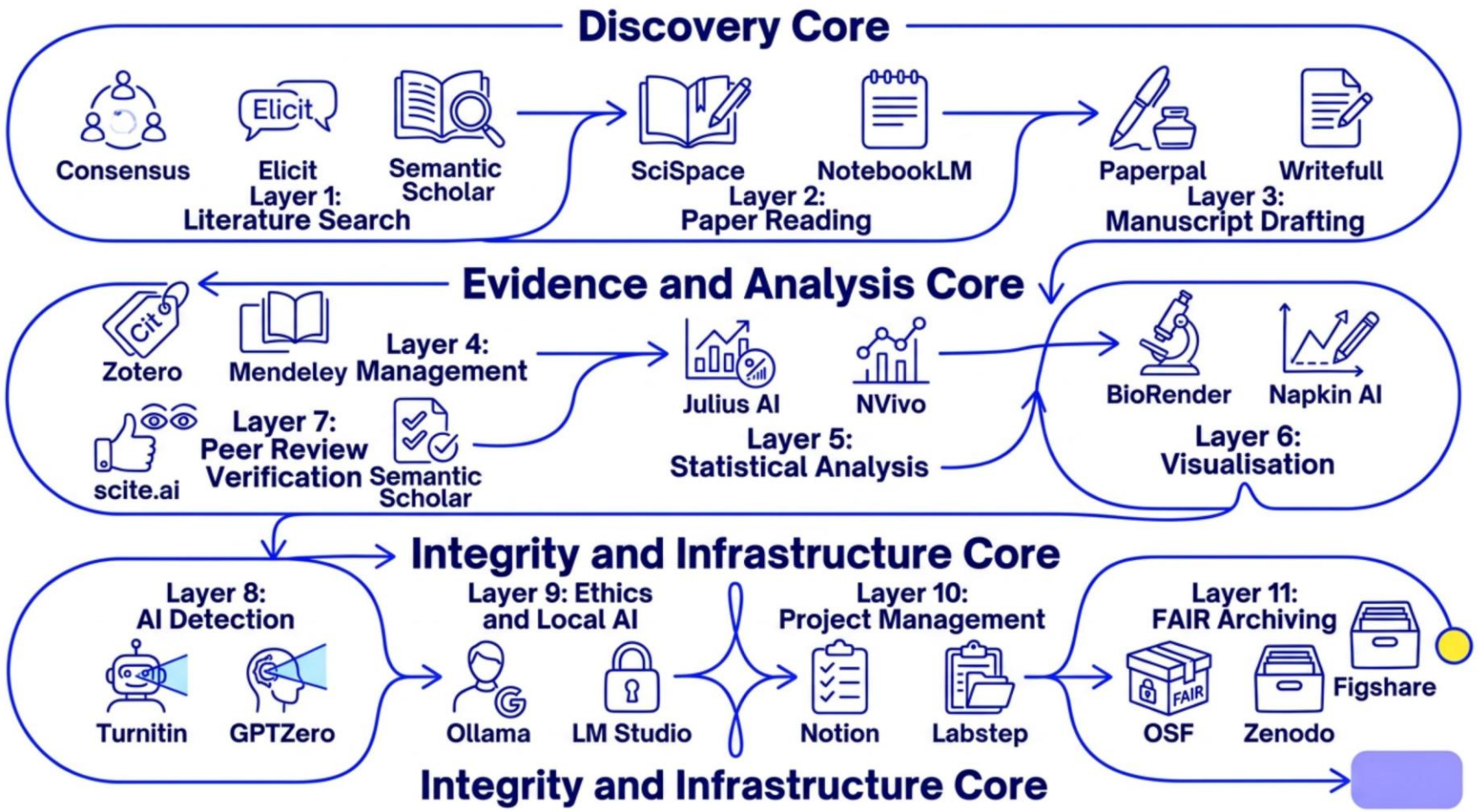
AlphaFold (protein structure), GNoME (materials)



LLMOps & Infrastructure

LangChain, LlamaIndex, LangSmith, Weights & Biases, local model runners

The Eleven-Layer Unified AI Research Framework



Key Ecosystem Concepts

Smart Citations (scite.ai)

Classifies each citing statement as **supporting, contrasting or mentioning** and flags retractions. Check whether a claim is supported or contested before relying on it.

FAIR Data

- **F**indable — persistent identifier, rich metadata
- **A**ccessible — standard protocols
- **I**nteroperable — standard formats
- **R**eusable — clear licence and provenance

Local AI — Why It Matters

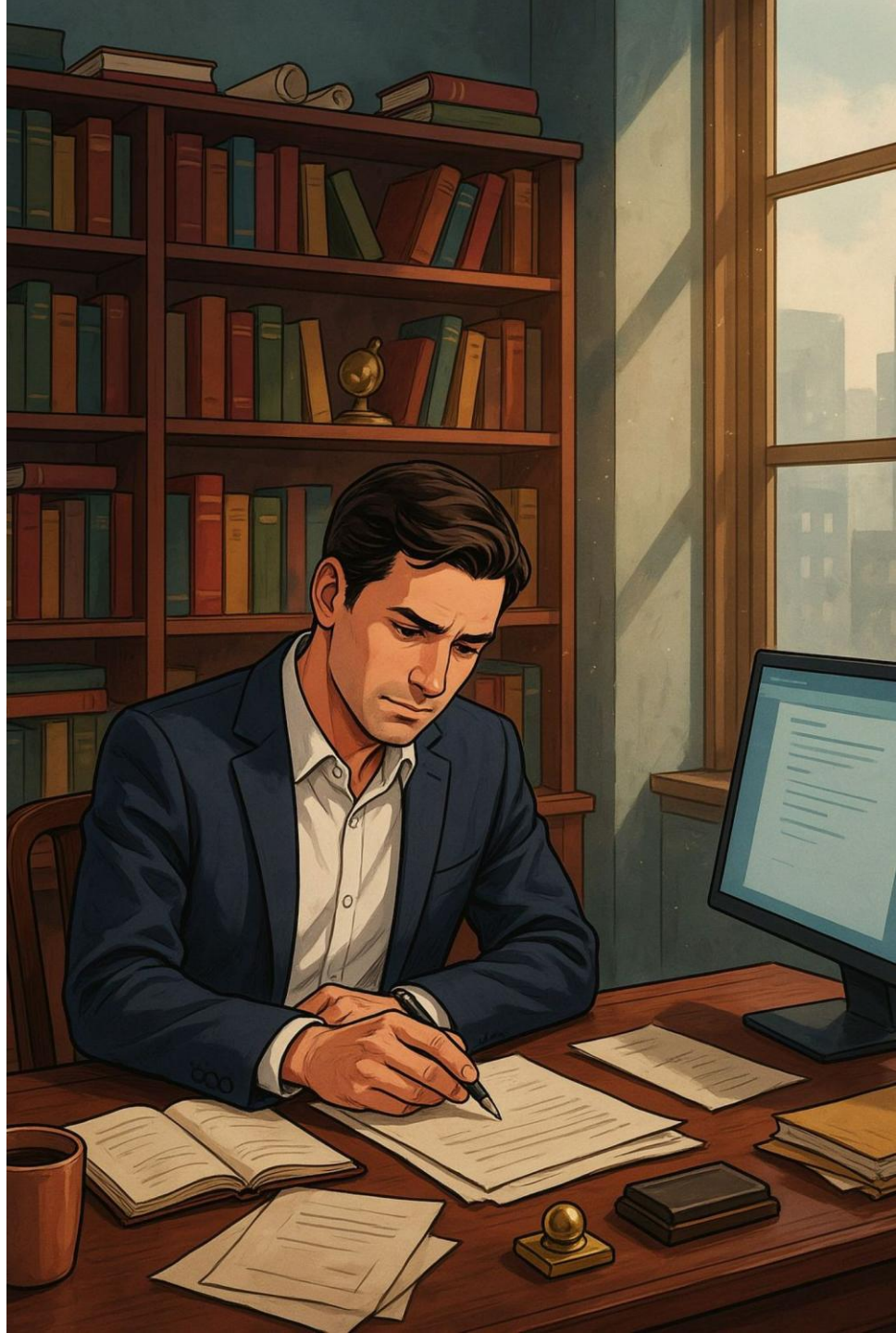
Sending patient data or interview transcripts to external servers can violate **GDPR, HIPAA, IRB rules** and India's **DPDP framework**. Tools: Ollama, LM Studio, LocalAI, llama.cpp — run quantised models offline.

MCP vs. A2A

- **MCP:** Connects AI apps to tools and data (agent → tool)
- **A2A:** Lets agents discover each other and delegate tasks (agent → agent)

PART A · SECTION 10

Risks, Ethics & Governance



Eight Challenges of AI-Assisted Research

1. Hallucination

Plausible but false content. Mitigate with RAG and mandatory citation checks.

2. Algorithmic Bias

Recommenders reinforce citation loops. Use multiple discovery strategies.

3. Bias & Fairness

Training data over-represents English, Western, high-income sources.

4. Lack of Explainability

Opaque models limit trust in high-stakes decisions.

5. Data Privacy

Conflicts with GDPR, HIPAA, ethics approvals. "Shadow AI" is an institutional risk.

6. Reproducibility

Random outputs and model version drift. Fix seeds, record versions, log prompts.

7. Over-Reliance

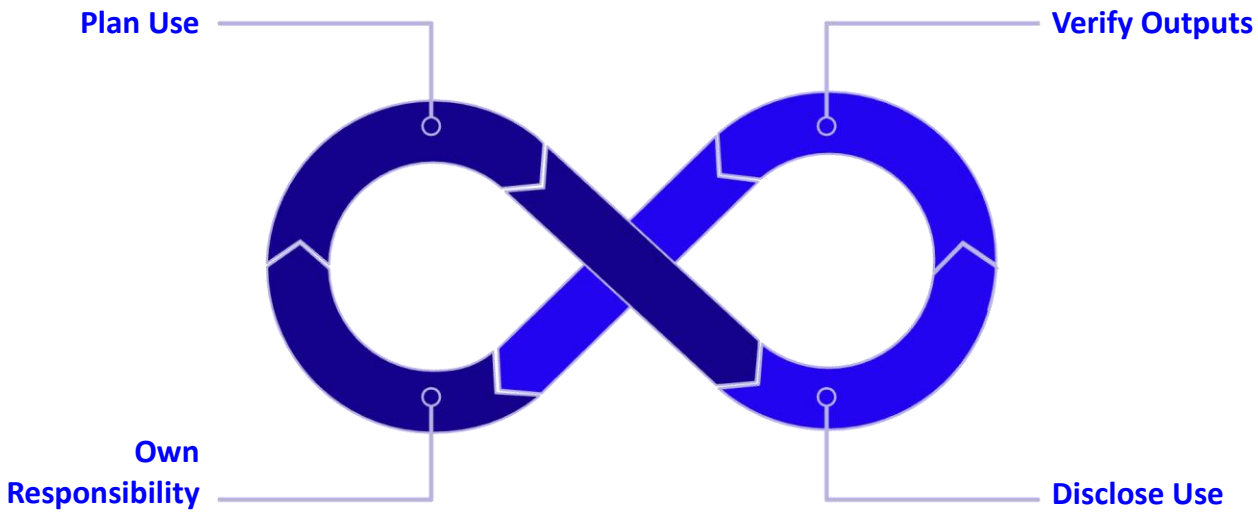
Cognitive atrophy: automation removes the struggle that builds expertise.

8. Stylistic Homogenisation

Everyone pushed toward the same prose, erasing scholarly voice.

Ethical AI Use & Disclosure

The Ethical AI Loop



What Disclosure Must Include

- Tool name and version
- Purpose of use
- Human checks performed
- Sections affected

Example Disclosure Statement

"We used an AI assistant (named, with version and date) to improve grammar in Sections 2 and 5 and to draft plotting code for Figure 3. The authors reviewed, tested and edited all output and take full responsibility. No confidential data were entered into AI tools."

⊗ AI cannot be listed as an author — authorship requires accountability, which only humans can bear.

Governance & Regulation

Four Governance Concerns

Authorship

Only humans can be accountable

Labour

Juniors may lose formative tasks

Environment

Energy, carbon and water costs

Dual Use

Autonomous agents need access controls

Regulatory Landscape

- **EU AI Act:** In force August 2024. High-risk duties postponed to December 2027 (stand-alone) and August 2028 (embedded in regulated products).
- **India:** Principles-based AI Governance Guidelines (November 2025). DPDP Rules require full compliance by **May 2027** — directly relevant to researchers handling personal data.

Key Takeaways

Verify Everything

AI output is an intermediate object — not evidence until you check it

Use Frameworks

R-C-S-E-O-V, UARF, Munzner's model — these are your exam anchors

Build Real Skills

Learn maths, code, read papers, build real systems, evaluate honestly

Own Your Thinking

Never outsource your thinking to AI — use AI to extend it

> "AI accelerates; researchers decide."

